

1 Paths and painting

Every way to build and paint a path. The theme changes only the paint, so every shape should keep its geometry and take the dark design's colours.

Fill rules: f and f*

ISO 32000-2 Table 59

swaps

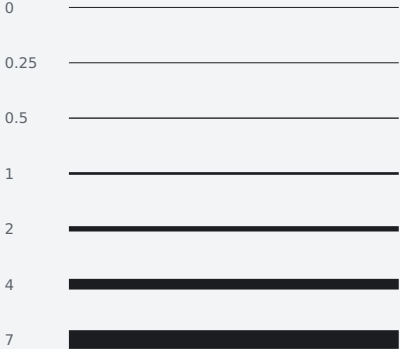


f nonzerof* even-odd

Line widths

ISO 32000-2 Table 56, T51

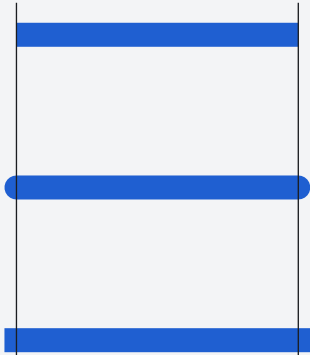
swaps



Line caps: J 0, 1, 2

ISO 32000-2 Table 53

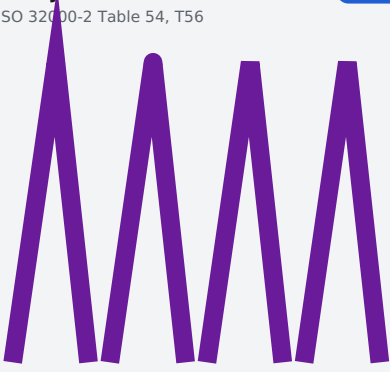
swaps



Line joins and miter limit

ISO 32000-2 Table 54, T56

swaps



j 0j 1j 2M 1.2

Dash patterns

ISO 32000-2 Table 55, T56

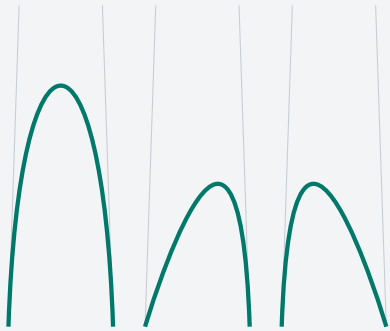
swaps



Curves: c, v and y

ISO 32000-2 Table 58

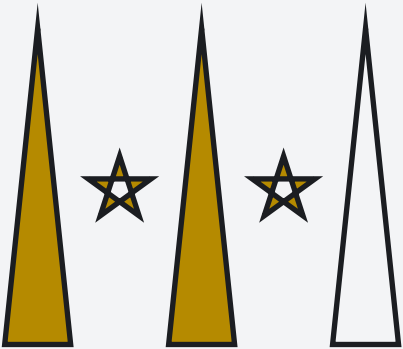
swaps



Fill and stroke: B, B*, b, b*, s

ISO 32000-2 Table 59

swaps



B B* b b* s

Clipping: W and W*

ISO 32000-2 Table 60

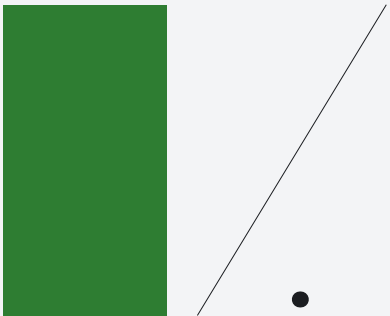
swaps



Rectangles, hairlines and dots

ISO 32000-2 Table 58, T51

swaps



Graphics state: q, Q, cm and gs

ISO 32000-2 Table 56, T57

swaps

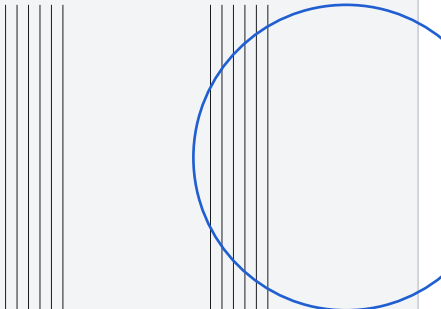


LW, LC, LJ, ML, D, RI and Font from one ExtGState

Stroke adjustment and flatness

ISO 32000-2 Table 57

viewer



SA and flatness are rendering hints

Strokes under a scaled CTM

ISO 32000-2 Table 56

swaps



PDF objects test for colour themes

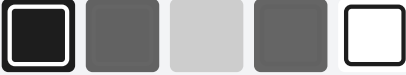
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2 Colour spaces

All eleven colour space families, fill and stroke. Each swatch is a fill with a stroked inner square. The dark build gives each space its own dark values.

DeviceGray: g, G
ISO 32000-2 Table 61, T73

swaps



DeviceRGB: rg, RG
ISO 32000-2 Table 61, T73

swaps



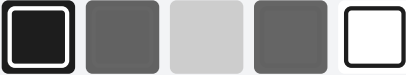
DeviceCMYK: k, K
ISO 32000-2 Table 61, T73

swaps



CalGray
ISO 32000-2 Table 62

swaps



CalRGB
ISO 32000-2 Table 63

swaps



Lab
ISO 32000-2 Table 64

swaps



ICCBased: sRGB
ISO 32000-2 Table 65, T66

swaps



ICCBased: Display P3
ISO 32000-2 Table 65, T67

swaps



ICCBased: CMYK (FOGRA39)
ISO 32000-2 Table 65, T68

swaps



Indexed over RGB, Lab and ICC
ISO 32000-2 Table 61

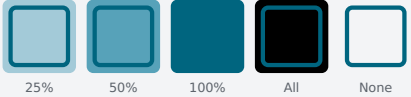
swaps



RGB RGB Lab Lab ICC

Separation: spot, All, None
ISO 32000-2 Table 61

swaps



25% 50% 100% All None

All = registration colour; None never marks

DeviceN and NChannel
ISO 32000-2 Table 70, T71

swaps



teal orange mix C+spot M+spot

Rendering intents: ri
ISO 32000-2 Table 69

viewer



Abs Rel Sat Perc

Intents change ICC conversion only

Default colour spaces in a form
ISO 32000-2 Table 34, T61

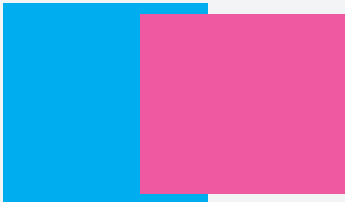
swaps



DefaultCMYK, DefaultRGB and DefaultGray remap k, rg and g

Overprint: OP, op and OPM
ISO 32000-2 Table 57, T146

viewer



Seen only with overprint preview

PDF objects test for colour themes

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3 Shadings and functions

All seven shading types, drawn with the sh operator, and all four function types. Mesh shadings carry colour at every vertex, so the dark build changes the mesh data.

Type 1: function-based
ISO 32000-2 Table 78, T42



Type 4 PostScript function

Type 2: axial
ISO 32000-2 Table 79, T40



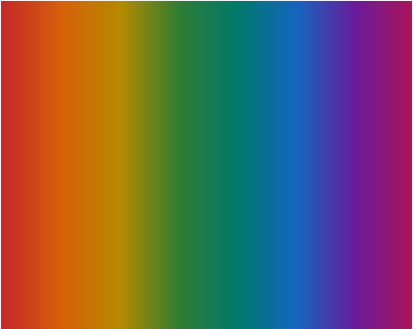
Type 2 exponential function, Extend

Type 3: radial
ISO 32000-2 Table 80, T41



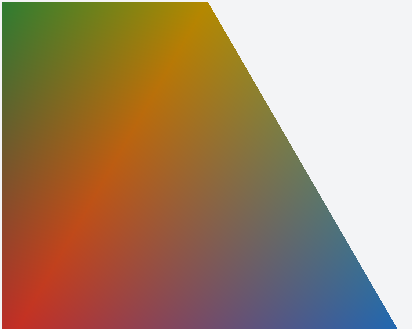
Type 3 stitching function

Axial with a sampled function
ISO 32000-2 Table 79, T39



Type 0 sampled function

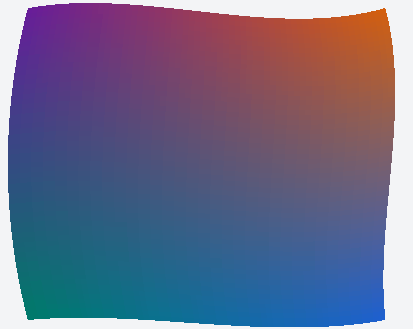
Type 4: free-form triangle mesh
ISO 32000-2 Table 81



Type 5: lattice-form mesh
ISO 32000-2 Table 82



Type 6: Coons patch mesh
ISO 32000-2 Table 83, T84



Type 7: tensor-product patch mesh
ISO 32000-2 Table 83, T85



Mesh with a function
ISO 32000-2 Table 81



Vertices carry t; a function maps t to colour

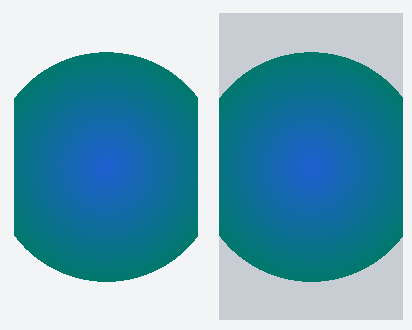
Axial shading in Lab
ISO 32000-2 Table 79, T64



Axial shading in DeviceN
ISO 32000-2 Table 79, T70



Background, BBox and AntiAlias
ISO 32000-2 Table 77



sh ignores Background; a pattern fill uses it

4 Patterns

Tiling patterns, coloured and uncoloured, and shading patterns for fills, strokes and text. A coloured pattern carries its colours inside; an uncoloured one takes them where it is used.

Tiling, coloured (PaintType 1)

swaps

ISO 32000-2 Table 74

Tiling, uncoloured (PaintType 2)

swaps

ISO 32000-2 Table 74

Colour given at use: [/Pattern /DeviceRGB]

Tiling types 1, 2, 3 and a Matrix

swaps

ISO 32000-2 Table 74

Shading pattern: fill

swaps

ISO 32000-2 Table 75

Shading pattern: stroke

swaps

ISO 32000-2 Table 75

Shading pattern: text

swaps

ISO 32000-2 Table 75, T104

Pattern

Pattern inside a pattern

swaps

ISO 32000-2 Table 74

Shading pattern with ExtGState

swaps

ISO 32000-2 Table 75, T57

Pattern's own ca 0.5

Uncoloured pattern over a spot colour

swaps

ISO 32000-2 Table 74, T61

[/Pattern /Separation ...]

5 Fonts

Every font type PDF defines: the standard 14 by reference, and embedded Type 1, TrueType, bare CFF, OpenType, two kinds of composite font and vertical writing. Text colour should follow the theme whatever the font type.

Standard 14 fonts, not embedded

swaps

ISO 32000-2 Table 108, T109

Times-Roman
Times-Bold
Times-Italic
Times-BoldItalic
Helvetica
Helvetica-Bold
Helvetica-Oblique

Helvetica-BoldOblique
Courier
Courier-Bold
Courier-Oblique
Courier-BoldOblique
αβγδπΩ
✓✓XX■★

The viewer supplies these fonts

Type 1, embedded (FontFile)

swaps

ISO 32000-2 Table 109, T124

Bitstream Charter: The quick brown fox jumps.
Bitstream Charter: The

TrueType, embedded (FontFile2)

swaps

ISO 32000-2 Table 109, T124

DejaVu Serif: The quick brown fox jumps.
DejaVu Serif: The qu

Bare CFF (FontFile3 /Type1C)

swaps

ISO 32000-2 Table 124, T125

Inter as CFF: The quick brown fox jumps.
Inter as CFF: The qu

OpenType CFF (FontFile3 /OpenType)

swaps

ISO 32000-2 Table 124, T125

Inter Bold OpenType: quick brown fox.
Inter Bold OpenTyp

Type 0 with CIDFontType2, Identity-H

swaps

ISO 32000-2 Table 115, T119

Ελληνικά αβγδε, Кириллица абвгд
Ελληνικά αβγδε,

Type 0 with CIDFontType0 (CFF CIDs)

swaps

ISO 32000-2 Table 115, T119

縦書きの日本語テキスト。横書きの文字列、字体見本。
縦書きの日本語テキスト。

Vertical writing: Identity-V

swaps

ISO 32000-2 Table 115, T116

縦書きの日本語
テキスト。横書

W2 and DW2 vertical metrics

Text state: Tc, Tw, Tz, Ts, TL, TJ

swaps

ISO 32000-2 Table 103, T106, T107

Tc spacing
Tw word spacing here
Tz 150
Rise: x² and TJ:A VA kerned
TD moves and sets the leading
the " operator sets Tw and Tc

Text matrix: Tm

swaps

ISO 32000-2 Table 106

Tm 0 Tm 15 Tm 30 Tm 45

6 Type 3 fonts and text effects

Type 3 glyphs that take the fill colour, glyphs that set their own colours, and a bitmap glyph; every text rendering mode, clipping to text, knockout and invisible text.

Type 3, uncoloured glyphs (d1)

ISO 32000-2 Table 110, T111

swaps

Type 3, coloured glyphs (d0)

ISO 32000-2 Table 110, T111

swaps

Colours live inside the glyph procedures

Type 3 glyph with a bitmap

ISO 32000-2 Table 110

stays

An emoji image: it stays

Rendering modes 0 to 3: Tr

ISO 32000-2 Table 104

swaps

Tr 0 Tr 1 Tr 2 Tr 3

3 is invisible

Rendering modes 4 to 7: clip

ISO 32000-2 Table 104

swaps

Tr 4 Tr 5 Tr 6 Tr 7

Text knockout: TK

ISO 32000-2 Table 57

swaps

TK false TK true

Invisible text over a scan

ISO 32000-2 Table 104

stays

Photo stays; Tr 3 text is invisible

Fill, stroke and pattern together

ISO 32000-2 Table 104, T75

swaps

Tiny, huge and scaled text

ISO 32000-2 Table 103

swaps

3 point text, still a colour

7 Images: colour spaces and bit depths

Photos stay the same in every theme. Diagrams stored as palette (Indexed) images swap only their colour table, so a dark theme recolours them without touching the pixels.

DeviceGray, 8-bit photo
ISO 32000-2 Table 87, T61

stays



DeviceGray at 1, 2 and 4 bits
ISO 32000-2 Table 87, T88

stays



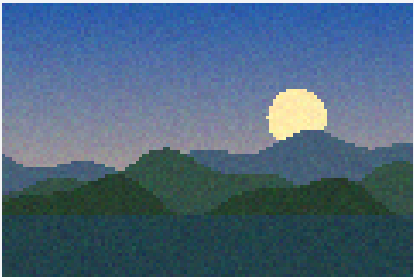
DeviceRGB, 8-bit photo
ISO 32000-2 Table 87

stays



DeviceRGB, 16-bit samples
ISO 32000-2 Table 87

stays



DeviceCMYK photo
ISO 32000-2 Table 87, T61

stays



Lab photo
ISO 32000-2 Table 87, T64

stays



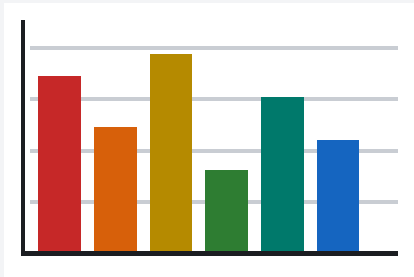
ICCBased: sRGB and Display P3
ISO 32000-2 Table 87, T65

stays



Indexed, 8-bit diagram
ISO 32000-2 Table 87, T61

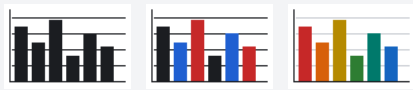
swaps



Only the colour table changes

Indexed at 1, 2 and 4 bits
ISO 32000-2 Table 87, T61

swaps



Indexed over Lab and ICC
ISO 32000-2 Table 87, T61

swaps



Separation image (spot ink)
ISO 32000-2 Table 87, T61

swaps



Tints of the teal spot ink

DeviceN image (two spot inks)
ISO 32000-2 Table 87, T70

swaps



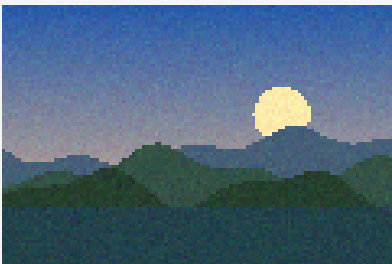
PDF objects test for colour themes

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8 Images: every filter

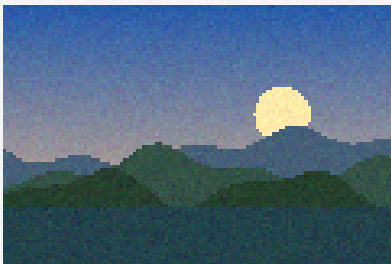
The same pictures compressed every way PDF allows. Filters change how pixels are stored, not what they look like, so each tile should look the same in every theme except the palette diagrams.

FlateDecode
ISO 32000-2 Table 6, T8



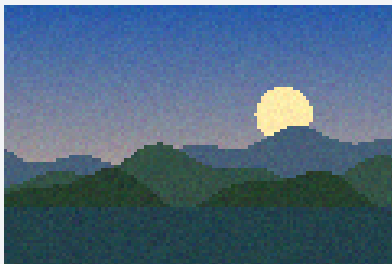
stays

FlateDecode with PNG predictor
ISO 32000-2 Table 8, T9, T10



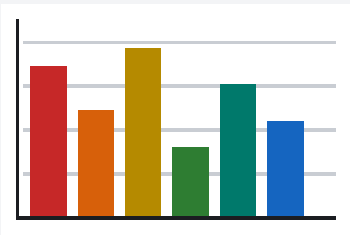
stays

LZWDecode
ISO 32000-2 Table 6, T7, T8



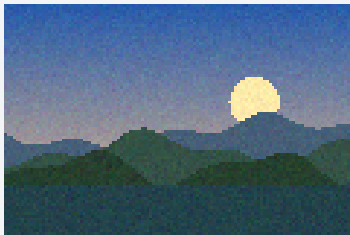
stays

RunLengthDecode
ISO 32000-2 Table 6



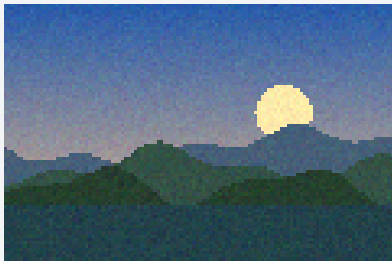
A palette diagram, so it swaps

ASCIIHexDecode then Flate
ISO 32000-2 Table 6



A filter chain

ASCII85Decode
ISO 32000-2 Table 6



stays

DCTDecode (JPEG), baseline
ISO 32000-2 Table 6, T13



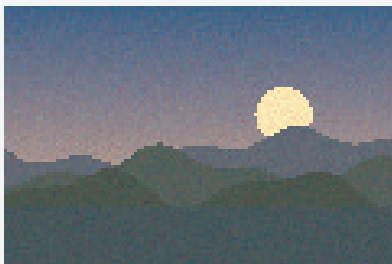
stays

DCTDecode, progressive greyscale
ISO 32000-2 Table 6, T13



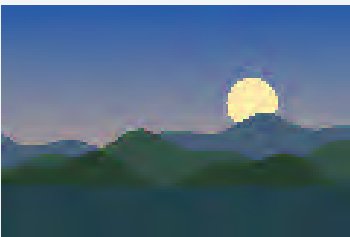
stays

DCTDecode, CMYK (Adobe)
ISO 32000-2 Table 6, T13, T88



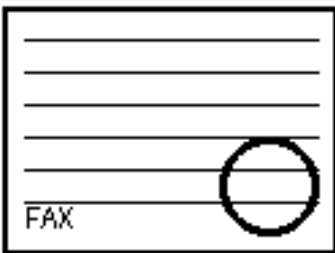
stays

JPXDecode (JPEG 2000)
ISO 32000-2 Table 6



Colour space comes from the codestream

CCITTFaxDecode, Group 4
ISO 32000-2 Table 6, T11



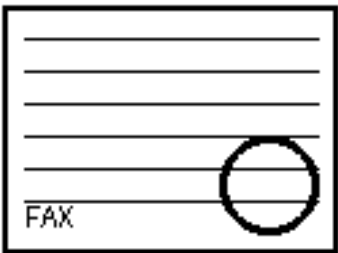
stays

CCITTFaxDecode, Group 3
ISO 32000-2 Table 6, T11



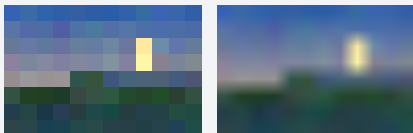
stays

JBIG2Decode (generic region, MMR)
ISO 32000-2 Table 6, T12



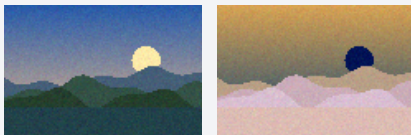
stays

Interpolate: false and true
ISO 32000-2 Table 87



stays

Alternates and a Decode array
ISO 32000-2 Table 87, T89, T88



Left has a print alternate; right is inverted by Decode

9 Masks and inline images

Stencil masks take the fill colour, so they swap. Masked and soft-masked photos stay. Inline images live inside the content stream; an inline palette image swaps through its named colour space.

Stencil mask (ImageMask)

swaps

ISO 32000-2 Table 87, T88



Painted in the current fill colour

Stencil mask with Decode [1 0]

swaps

ISO 32000-2 Table 88

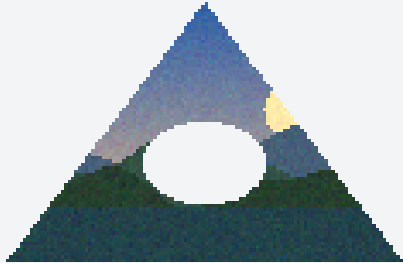


The same mask, inverted

Explicit mask (Mask stream)

stays

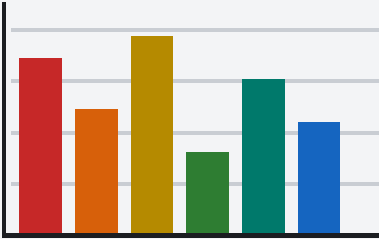
ISO 32000-2 Table 87



Colour-key mask (Mask array)

swaps

ISO 32000-2 Table 87

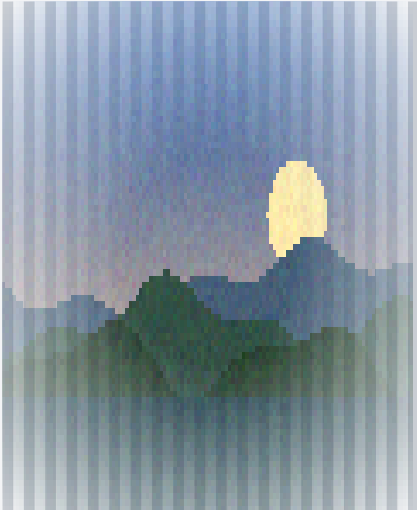


Index 0 (paper) is keyed out

Soft mask (SMask)

stays

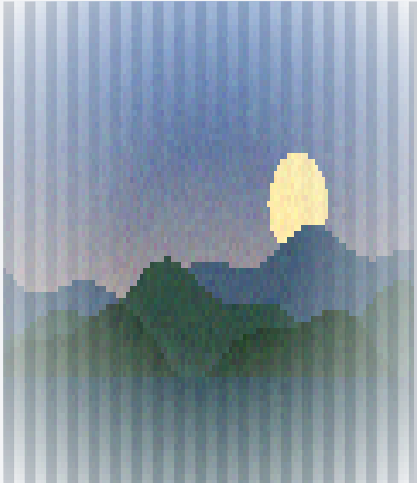
ISO 32000-2 Table 87, T143



Soft mask with Matte

stays

ISO 32000-2 Table 144



Colours premultiplied against white

JPEG 2000 with SMaskInData

stays

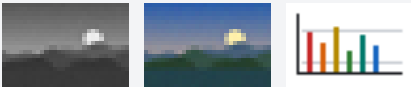
ISO 32000-2 Table 87



Inline images: grey, RGB, Indexed

swaps

ISO 32000-2 Table 90, T91, T92

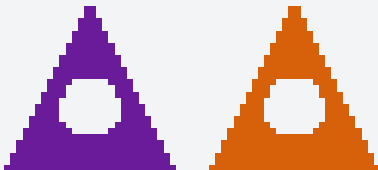


Indexed swaps through a named space; grey and RGB stay

Inline stencil mask

swaps

ISO 32000-2 Table 91



PDF objects test for colour themes

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10 Transparency: opacity and blend modes

Constant opacity and all sixteen blend modes. Blending mixes the theme's colours, so a mode that reads well on light paper, such as Multiply, can vanish on dark paper.

Fill opacity: ca

ISO 32000-2 Table 57, T136

swaps

Stroke opacity: CA

ISO 32000-2 Table 57

swaps

Alpha is shape: AIS

ISO 32000-2 Table 57

swaps

Left AIS false, right AIS true

All sixteen blend modes: BM

ISO 32000-2 Table 134, T135

swaps

Normal

Multiply

Screen

Overlay

Darken

Lighten

ColorDodge

ColorBurn

HardLight

SoftLight

Difference

Exclusion

Hue

Saturation

Color

Luminosity

A blue disc blended over a two-colour gradient, one mode per cell

Highlighter: Multiply, or Screen in dark

ISO 32000-2 Table 134

swaps

Highlighted words

Multiply hides a highlight on dark paper, so the dark build sw

Screen and Lighten on paper

ISO 32000-2 Table 134

swaps

Difference and Exclusion on text

ISO 32000-2 Table 134

swaps

Difference over text

Exclusion over text

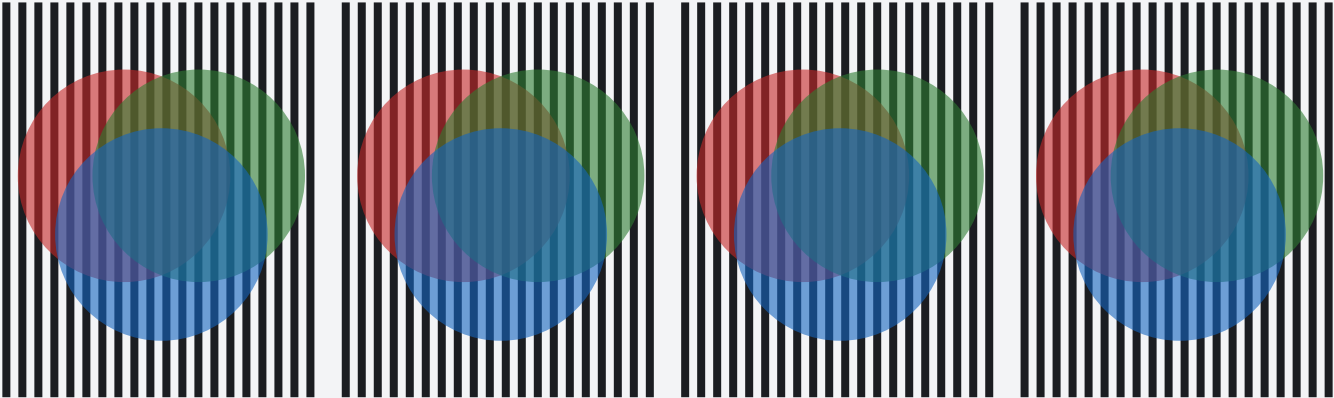
11 Transparency: groups and soft masks

Isolated and knockout groups, a group colour space, and soft masks. A luminosity mask is made of greys that mean opacity, not colour, so the theme must leave the mask alone and swap only what it masks.

Isolated and knockout groups: I and K

ISO 32000-2 Table 94, T145

swaps



I false, K false

I true, K false

I false, K true

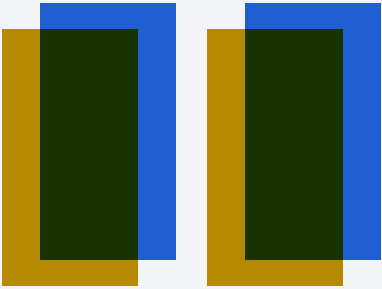
I true, K true

Three translucent discs in a group over stripes; each cell changes I and K

Group colour space: CS

ISO 32000-2 Table 145

swaps

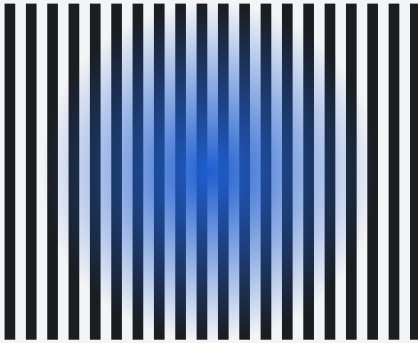


Multiply blended in RGB, then in CMYK

Luminosity soft mask with BC

ISO 32000-2 Table 142

swaps

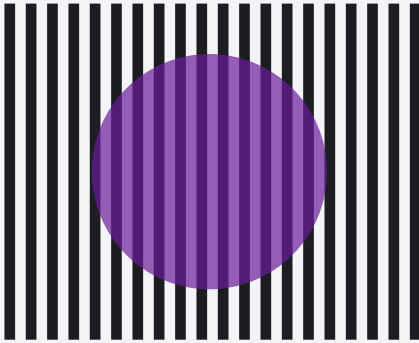


Mask greys stay; the masked colour swaps; BC black outside

Alpha soft mask

ISO 32000-2 Table 142

swaps

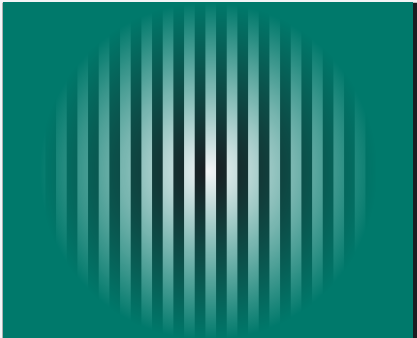


Mask from a group's opacity

Soft mask with a transfer function

ISO 32000-2 Table 142

swaps

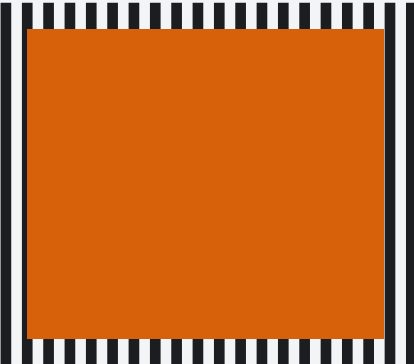


TR inverts the mask

SMask None resets the mask

ISO 32000-2 Table 57

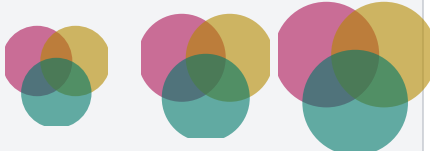
swaps



Transparency group reused

ISO 32000-2 Table 94

swaps



One group form drawn three times

PDF objects test for colour themes

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12 Form XObjects

Reusable drawings: placed many times, nested, clipped by their BBox, and drawn in the colour of the page that calls them. A theme that swaps a form must swap it everywhere it is used.

One form, placed four times

ISO 32000-2 Table 93, T86

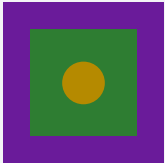
swaps



Nested forms, three deep

ISO 32000-2 Table 93

swaps



Form drawn in the caller's colour

ISO 32000-2 Table 93

swaps



No colour inside the form

BBox clips and Matrix maps

ISO 32000-2 Table 93

swaps

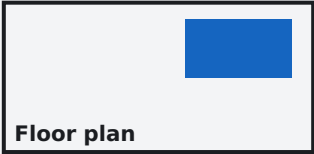


The circle is cut by the form's BBox

Form with Measure and PtData (PDF 2.0)

ISO 32000-2 Table 93, T266, T272

swaps



Floor plan

A scaled drawing: 1 pt in the form is 1 cm

Reference XObject (Ref)

ISO 32000-2 Table 93, T95

swaps



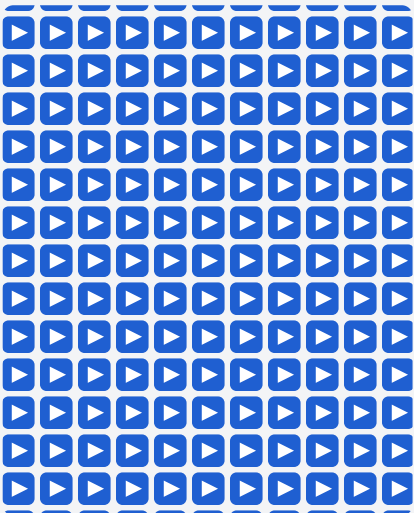
Proxy for Ref

Viewers draw the proxy content

Form in a tiling pattern cell

ISO 32000-2 Table 74, T93

swaps



Form with its own Group and Matrix

ISO 32000-2 Table 93, T94

swaps

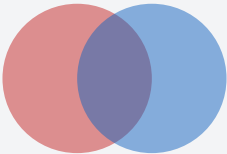
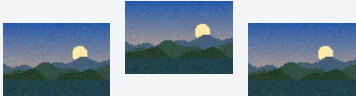


Image inside a form, reused

ISO 32000-2 Table 93, T87

stays



13 Optional content (layers)

Layers that are on, off, view-only, print-only, zoom-dependent, language-dependent, locked and in radio groups, plus a visibility expression. Click the labels to toggle layers in viewers that support SetOCGState. Content in a layer swaps like any other content.

Default on

ISO 32000-2 Table 96, T99

swaps

Shown by default

Default off

ISO 32000-2 Table 96, T99

swaps

Hidden by default

Off until toggled

View and print usage

ISO 32000-2 Table 100, T101

swaps

Screen only

AS sets it from Usage

Print-only layer

ISO 32000-2 Table 100, T101

swaps

Hidden on screen

Zoom-dependent layer

ISO 32000-2 Table 100

swaps

Zoom 150%+

Shown at 150% zoom and above

Language layers

ISO 32000-2 Table 100

swaps

English

Français

Radio-button layers: RBGroups

ISO 32000-2 Table 99

swaps

C

Only one of A, B, C at a time

Visibility expression: OCMD VE

ISO 32000-2 Table 97

swaps

Shown when on AND NOT off

Locked layer, layered annotation and form

ISO 32000-2 Table 99, T166, T93

swaps

The square annotation and the form both carry OC

14 Annotations: notes, shapes and lines

Each annotation carries an appearance stream drawn in the theme's colours, and its colour entries (C, IC, DA, DS, RC) say the same thing, for viewers that redraw it. The tiles marked viewer have no appearance stream, so the viewer draws them from those entries.

Text (sticky note) with appearance
ISO 32000-2 Table 175, T186



Has a closed popup

Review state and a reply: IRT, RT, State
ISO 32000-2 Table 172, T174, T175



Reply to the note

Review state: Accepted (NoView)

Marked state (NoView)

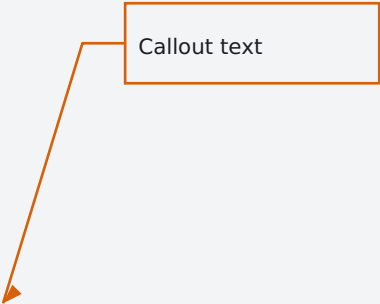
A reply thread on the sticky note; the state is hidden

FreeText with DA, DS and RC
ISO 32000-2 Table 177

Free text, drawn in ink
and one red phrase

Rich text colours in RC

FreeText callout: CL, LE, IT
ISO 32000-2 Table 177, T179



FreeText typewriter
ISO 32000-2 Table 177

Typed on the page

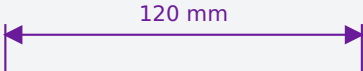
No border, text only

Line: all ten line endings
ISO 32000-2 Table 178, T179



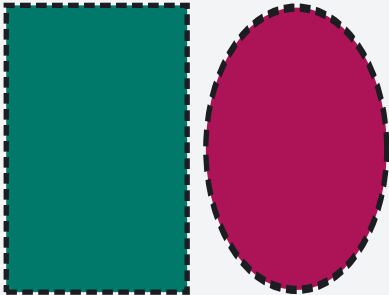
Closed endings filled with IC

Line dimension with Measure
ISO 32000-2 Table 178, T266, T267



Leader lines and a caption

Square and Circle with IC
ISO 32000-2 Table 180, T168

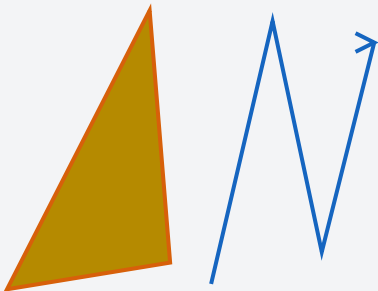


Dashed border style

Cloudy border: BE
ISO 32000-2 Table 169



Polygon and PolyLine
ISO 32000-2 Table 181

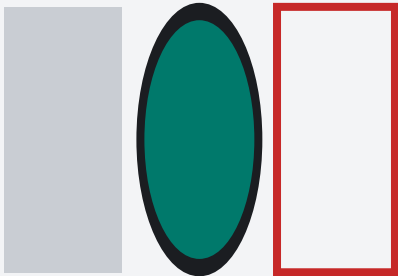


Cloud intent; open path with endings

Ink: InkList
ISO 32000-2 Table 185



C and IC in grey, CMYK and none
ISO 32000-2 Table 166, T180



C [] is transparent: no border is drawn

PDF objects test for colour themes

page 14

15 Annotations: text markup, stamps and the rest

Text markup over real text, stamps, a popup, a redaction mark, attachments, a watermark, printer's marks and a trap network, plus the PDF 2.0 annotation blend mode and fill opacity.

Highlight

ISO 32000-2 Table 182

swaps

Highlight this phrase

Multiply in light; the dark build uses Screen

Underline, Squiggly, StrikeOut

ISO 32000-2 Table 182

swaps

Underline this text

Squiggly this text

StrikeOut this text

Caret: Sy P and None

ISO 32000-2 Table 183

swaps

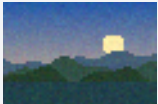
Insert here and here

Stamp: drawn and image

ISO 32000-2 Table 184

swaps

APPROVED



The image stamp stays

Stamp names, with appearances

ISO 32000-2 Table 184

swaps

DRAFT

CONFIDENTIAL

FINAL

FOR COMMENT

TOP SECRET

EXPIRED

The legacy file has the same stamps without appearances

Popup, open

ISO 32000-2 Table 186

viewer



Popups are drawn by the viewer

Redact (marked, not applied)

ISO 32000-2 Table 195

swaps

Account1234 5678

IC and OverlayText apply once redacted

FileAttachment: six icon names

ISO 32000-2 Table 187, T43

swaps



Each has an appearance; the legacy file leaves them to the viewer

Watermark with FixedPrint

ISO 32000-2 Table 193, T194

swaps

DRAFT

PrinterMark: colour bar and target

ISO 32000-2 Table 398, T399

stays

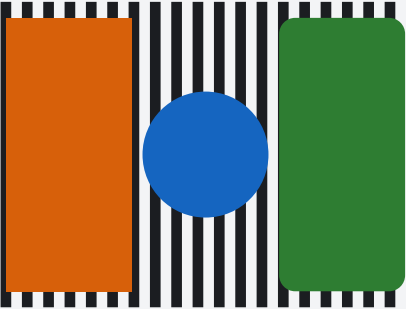


Print production marks keep their true colours

Annotation BM and ca (PDF 2.0), R and D

ISO 32000-2 Table 166, T170

swaps



Multiply square; half-opaque circle; hover and press change

TrapNet (deprecated in PDF 2.0)

ISO 32000-2 Table 403, T404

stays



Print trapping; keeps its colours

16 Links and actions

Link borders are drawn by the viewer from the C entry, so C must follow the theme. Actions that show, hide or recolour things on the page are here too; colours set by JavaScript are out of a theme's reach, which the spec should say.

Link with a border colour: C

viewer

ISO 32000-2 Table 176, T210

example.com

URI action; the viewer draws the border from C

Highlight modes: H

viewer

ISO 32000-2 Table 176

H /N H /I H /O H /P

N, I, O and P change how a click looks

Link over two lines: QuadPoints

viewer

ISO 32000-2 Table 176

This link runs over two lines of text.

Link with a dashed border style: BS

viewer

ISO 32000-2 Table 168, T176

Dashed border link

Destinations: XYZ, Fit, FitH, FitR, FitB

swaps

ISO 32000-2 Table 149

XYZ to page 3

Fit to page 4

FitH to page 5

FitR to page 6

FitB to page 7

Named, remote and embedded go-to

swaps

ISO 32000-2 Table 202, T203, T204, T205

Named destination

Remote file (GoToR)

Embedded file (GoToE)

GoToR names a file that is not shipped

Named actions

swaps

ISO 32000-2 Table 215, T216

FirstPage

PrevPage

NextPage

LastPage

Hide action and the Hidden flag

swaps

ISO 32000-2 Table 214, T167

Show it

Hide it

Show and hide a hidden square

JavaScript action and Next

viewer

ISO 32000-2 Table 221, T196

Make the name field yellow

Reset its colour

Colours in scripts are out of a theme's reach

Launch, Thread and Transition actions

viewer

ISO 32000-2 Table 207, T209, T219

Launch readme.txt

Read the article thread

Dissolve transition

Launch is usually blocked; Thread follows the article

Flags: NoView, Print off, ToggleNoView

viewer

ISO 32000-2 Table 167

Left prints but is not shown; middle shows but does not print

Page and document additional actions

viewer

ISO 32000-2 Table 197, T198, T200

Hover or click here

This page logs when opened and closed

17 Form fields: text and choice

Text and choice fields. Each widget's appearance is drawn in the theme's colours, and its MK background and border and its DA text colour say the same, because viewers redraw fields as soon as someone types.

Text field

ISO 32000-2 Table 226, T228, T232

swaps

Ada Lovelace

Multiline text field

ISO 32000-2 Table 231

swaps

Several lines of text,
wrapped by the field,
in the theme's ink.

Password field

ISO 32000-2 Table 231

swaps

.....

Comb field: MaxLen 6

ISO 32000-2 Table 231, T232

swaps

A B 1 2 C D

File-select field

ISO 32000-2 Table 231

swaps

report.pdf

Rich text field: RV and DS

ISO 32000-2 Table 228, T231

swaps

Plain and coloured text

Span colours live in RV

Alignment: Q 1 and Q 2

ISO 32000-2 Table 228

swaps

Centred

Right aligned

Border styles: beveled, inset, underline

ISO 32000-2 Table 168, T192

swaps

Beveled

Inset

Underline

Bevel shades come from MK BG

Required and read-only flags

ISO 32000-2 Table 227

swaps

Required

Read only

Number format script: red negatives

ISO 32000-2 Table 199, T221

viewer

-£42.50

AFNumber_Format sets color.red when negative

Calculated field: CO

ISO 32000-2 Table 199, T224

viewer

20 42 62

Total = a + b, by script

List box

ISO 32000-2 Table 233, T234

swaps

Red

Green

Blue

Purple

The selected row is drawn in the accent

Multi-select list box

ISO 32000-2 Table 233, T234

swaps

Red

Green

Blue

Purple

Combo box

ISO 32000-2 Table 233

swaps

Medium

Editable combo box

ISO 32000-2 Table 233

swaps

Bristol

18 Form fields: buttons and signatures

Check boxes in all six styles, a radio group, push buttons with captions, rollover and icons, a signature field, and scripts that change colours on focus. A theme can recolour appearances; it cannot reach a colour written in a script.

Check boxes: six styles

ISO 32000-2 Table 229, T230, T192

swaps

☒

check

☐

circle

☒

cross

☐

diamond

☒

square

☐

star

MK CA picks the ZapfDingbats mark

Radio group

ISO 32000-2 Table 229, T230

swaps

☐

S

☒

M

☐

L

One parent field, three widgets

Push button: N, R and D appearances

ISO 32000-2 Table 229, T170, T192, T241

swaps

Reset the form

Captions CA, RC and AC; resets the form

Icon buttons: MK I, RI, IX, IF and TP

ISO 32000-2 Table 192, T250

swaps

Play

Icon only, and caption below; hover and press swap the icon

Submit button

ISO 32000-2 Table 239, T240

viewer

Submit

Submits to an address that does not exist

Signature field, unsigned

ISO 32000-2 Table 235, T236, T237

swaps

Sign here

Lock and seed value dictionaries

Focus and blur scripts recolour a field

ISO 32000-2 Table 197, T221

viewer

Click into me

Fo sets a yellow fill; BI removes it

Hidden field shown by a button

ISO 32000-2 Table 167, T214

swaps

Show the field

Field hierarchy: a.b names

ISO 32000-2 Table 226

swaps

1 High Street

Bristol

Parent 'address' with two children

Read-only check box and NoExport

ISO 32000-2 Table 227

swaps

☒

Read only, not exported

Import data action

ISO 32000-2 Table 243

viewer

Import values

Imports field values from an FDF file

MK colours in grey and CMYK

ISO 32000-2 Table 192

swaps

Grey BG and BC

CMYK BG and BC

No BG (transparent)

BG and BC can have 1 or 4 components, or none

PDF objects test for colour themes

page 18

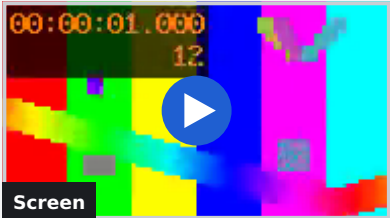
19 Multimedia and 3D

Video, sound and a 3D model. The media themselves stay as they are, but their posters and frames, the floating window's background (B), and the 3D view's background, render mode, cross-section and measurement colours all belong to the theme.

Screen with a Rendition action

swaps

ISO 32000-2 Table 190, T218, T282, T293



MediaScreenParams B is the window background

Rendition actions: play, pause, stop

viewer

ISO 32000-2 Table 218

Play the video

Pause

Resume

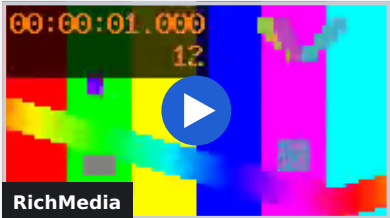
Stop

RichMedia (video)

swaps

ISO 32000-2 Table 333, T334, T335, T338, T339, T340, T341, T342, T222

RichMediaExecute: play



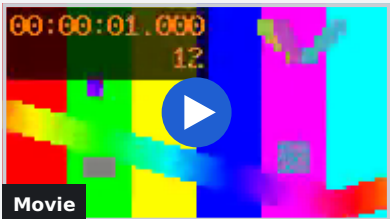
Opens in a floating window when activated

Movie annotation (deprecated)

swaps

ISO 32000-2 Table 189, T306, T307, T213

Movie action: play



Movie action plays it

Sound annotation and action (deprecated)

swaps

ISO 32000-2 Table 188, T305, T212

 Sound action: play

A 440 Hz tone; the speaker icon swaps

3D annotation (U3D)

swaps

ISO 32000-2 Table 309, T310, T311, T315, T317, T318, T320, T322, T323, T326-T331



3D model: click to activate

3 views: BG, RM, LS, a section, a node, 5 measures

GoTo3DView actions, 3D markup

viewer

ISO 32000-2 Table 220, T173, T324

Default view

Wireframe view

Cross-section view

Switch views; the note at top right comments on view 1

Projection annotation (PDF 2.0)

swaps

ISO 32000-2 Table 172, T332, T327



Shows the 3D measurement on the page

Selector rendition with MH and BE

viewer

ISO 32000-2 Table 277, T278, T279, T283

Play through a selector

Picks the first rendition the viewer can play

Floating window colour

swaps

ISO 32000-2 Table 293, T294, T295



B = paper colour

B is the background of the media window

Embedded media files

stays

ISO 32000-2 Table 43, T44, T45

clip.mp4, 12.8 kB

tone.wav, 4.8 kB

The MP4 and the tone are in the EmbeddedFiles tree

What viewers show

viewer

Most viewers show only the posters here. Acrobat plays the video and the 3D model; pdf.js, Poppler, MuPDF and PDFium draw the posters and skip the media.

20 Page boxes, thumbnails and navigation

This page has a bleed: its MediaBox is 18 pt larger than its TrimBox on every side. Box guide colours (BoxColorInfo), the thumbnail, transitions, article threads, measurement viewports and navigation steps are all here; most show only in a viewer's own panels or modes.

Page boxes and BoxColorInfo

ISO 32000-2 Table 31, T396, T397

CropBox

BleedBox

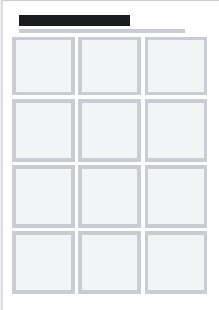
TrimBox

ArtBox

Box guide colours are viewer chrome

Thumbnail: Thumb

ISO 32000-2 Table 31



An Indexed image, so its table swaps

Transition and display time

ISO 32000-2 Table 31, T164

Trans: Wipe, 0.8 s, from the bottom. Dur: 30 s before the next page in full-screen mode.

Seen in full-screen mode

Article thread: beads

ISO 32000-2 Table 162, T163

Bead one: an article starts here and continues below.

Bead two: then on to the rotated page.

Two beads here, two on the next page

Viewport with a scale: Measure RL

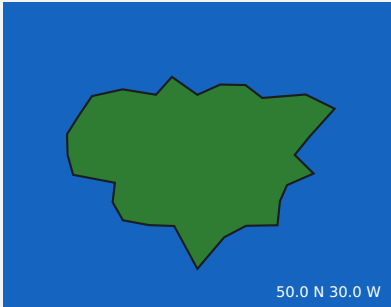
ISO 32000-2 Table 265, T266, T267, T268



Measuring tools read 1 cm on the bar as 1 m

Geospatial viewport: GEO, PROJCS, PtData

ISO 32000-2 Table 265, T269, T270, T271, T272



A made-up island at 50 N, 30 W; map colours swap

Navigation steps: PresSteps

ISO 32000-2 Table 165

Step one shows the hidden layer on page 13; step back hides it again.

Arrow keys step through layers

Document parts and GoToDp

ISO 32000-2 Table 206, T408, T409

[Go to the appendix part](#)

Parts: content pages and appendix

Associated file on a page: AF

ISO 32000-2 Table 31, T43, T45

This page carries a CSV of its own tiles, with AFRelationship Supplement.

Outlines with colour and style

ISO 32000-2 Table 151, T152

> Plain

> Italic

> Bold

> Bold italic

Bookmark colours are shown in the panel

Page labels

ISO 32000-2 Table 161

The appendix pages are labelled A-1, A-2 and so on. Viewers show the label in the page box.

Pages show as 1 to n, then A-1 on

Page output intent, by reference

ISO 32000-2 Table 31, T401, T402

This page's output intent names FOGRA39 by reference instead of embedding the profile.

PDF 2.0: a page names its own condition, here by profile reference

PDF objects test for colour themes

page 20

21 Rotated page: Rotate 90

The page is stored upright and turned a quarter turn by the viewer. Annotations with NoRotate stay upright, and NoZoom keeps their size.

Content on a rotated page

ISO 32000-2 Table 31

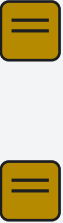
swaps



NoRotate and NoZoom flags

ISO 32000-2 Table 167

viewer



Article thread continues

ISO 32000-2 Table 163

viewer

Bead three: the article continues on this rotated page and ends in the next box.

Last bead

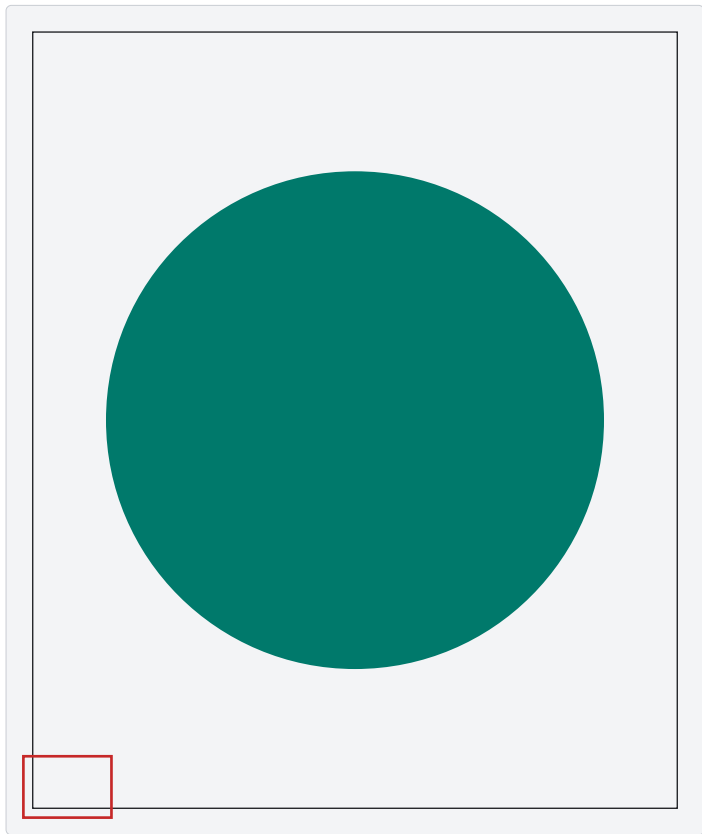
ISO 32000-2 Table 163

viewer

Bead four: the end of the article.

22 UserUnit 2

This page's MediaBox is half of A4, and UserUnit 2 makes each unit 1/36 inch, so it shows at A4 size. Content, annotations and line widths all scale.



23 Tagged structure and colour attributes

Reflow, read-aloud and PDF-to-HTML tools use the structure tree, not the page. Its Layout attributes carry colours of their own, which a theme has to swap with the page.

Table with Layout colours

ISO 32000-2 Table 371, T377, T378, T384

Theme	Paper	Ink
Light	#FFFFFF	#1B1D21
Dark	#18191C	#E8EAED
Tinted	#FDF6E3	#1B1D21

TH BackgroundColor and Color, table BorderColor

List with numbering

ISO 32000-2 Table 370, T382

1. Swap palettes

2. Paint the paper

3. Draw as usual

Lbl Color differs from LBody

Underline with TextDecorationColor

ISO 32000-2 Table 380

A decorated phrase

Attribute class: ClassMap and C

ISO 32000-2 Table 354, T355, T360

Warning: the colours here come from a class.

Class 'Warning' sets two colours

FENote, E and Lang (PDF 2.0 namespace)

ISO 32000-2 Table 355, T356, T366, T368

PDF themes need a note¹

Le mode sombre, en français.

1 Footnote text, also themed.

Formula with a MathML file

ISO 32000-2 Table 374, T43

$L = 0.2126 R + 0.7152 G + 0.0722 B$

Alt text and AF: Supplement

Ruby and Warichu

ISO 32000-2 Table 369

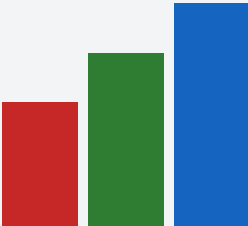
かんじ
漢字

(割注
にほんご)

Readings drawn above the base text

Figure with BBox and Caption

ISO 32000-2 Table 373, T372, T379



Colour spaces, p2
Images, p7
Annotations, p14

Figure 1: three bars

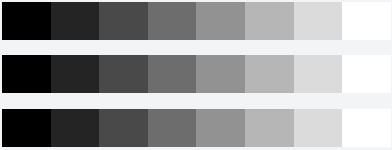
24 Printing and device-dependent objects

Halftones, transfer functions, black generation and the rest act when a page is printed or converted for a device. A theme changes the colours that go in; these still apply on the way out, so the spec should say whether a theme applies before or after them. The deprecated objects are here because old files still carry them.

Halftone type 1: spot functions

ISO 32000-2 Table 126, T127, T128

stays



Print only

Halftone type 5 per colorant

ISO 32000-2 Table 132

stays

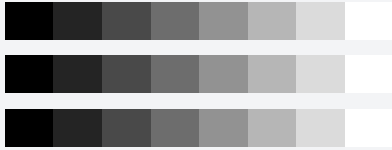


Different screens for C, M, Y, K

Threshold halftones: types 6, 10, 16

ISO 32000-2 Table 129, T130, T131

stays



Transfer functions: TR and TR2

ISO 32000-2 Table 57

viewer



Some viewers apply TR, so it changes what the theme draws

Black generation and undercolour removal

ISO 32000-2 Table 57

stays



BG, BG2, UCR, UCR2: RGB to CMYK at print time

SM, HTO and UseBlackPtComp

ISO 32000-2 Table 57

stays



Smoothness; PDF 2.0 halftone origin and BPC

SeparationInfo

ISO 32000-2 Table 400

stays



Says this page is one plate of a separated job

Document output intent

ISO 32000-2 Table 29, T401

stays

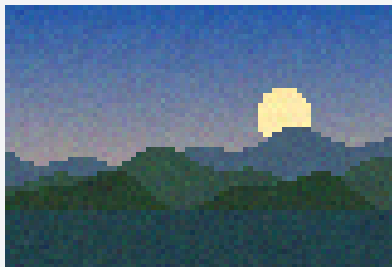
Device colours mean FOGRA39 here, so a theme's DeviceCMYK values are read through this profile.

FOGRA39 profile embedded

OPI 1.3 and 2.0 on an image (deprecated)

ISO 32000-2 Table 405, T406, T407

stays



MType1 font (deprecated)

ISO 32000-2 Table 108, T109

swaps

Multiple master text

Not embedded; the viewer substitutes

BX/EX and marked points: MP, DP

ISO 32000-2 Table 33, T352

swaps



An unknown operator inside BX/EX is skipped

Not tested: removed or invisible

stays

PostScript XObjects and the PS operator (removed in PDF 2.0), XFA forms (deprecated), Web Capture and alternate presentations: none of these draw anything a theme can reach.

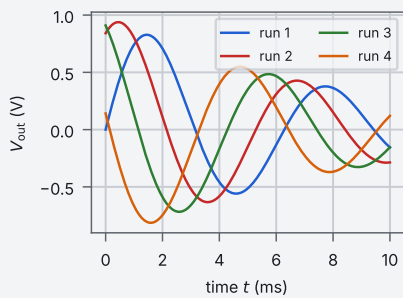
25 Scientific figures: matplotlib

Figures made by matplotlib 3.11 with its defaults, from one script run with light and dark colours. Fonts are Type 3, every figure starts with a background rectangle, markers are one XObject drawn thousands of times, and heatmaps and colour bars are Indexed images. Data colour maps that work on both papers stay; a diverging map swaps to a dark-midpoint one.

Line plot: series, legend, grid

ISO 32000-2 Table 59, T110

swaps

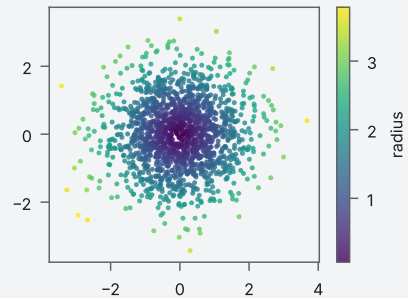


Type 3 fonts, background rectangle

Scatter coloured by value

ISO 32000-2 Table 86, T93, T87

swaps

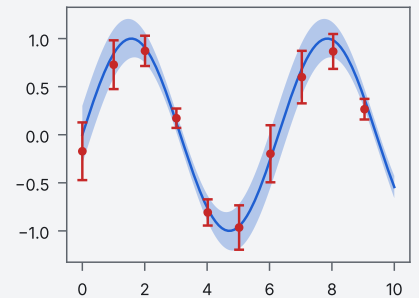


1,500 Do of one marker; data colours stay

Error bars and a confidence band

ISO 32000-2 Table 57, T86

swaps

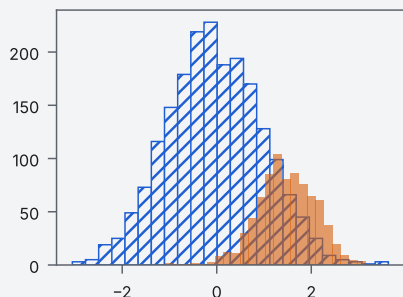


Band at 30% opacity

Histogram with hatching

ISO 32000-2 Table 74, T57

swaps

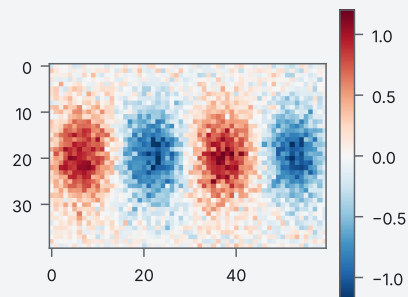


Hatch colour lives inside the pattern

Heatmap, diverging map

ISO 32000-2 Table 87, T61

swaps

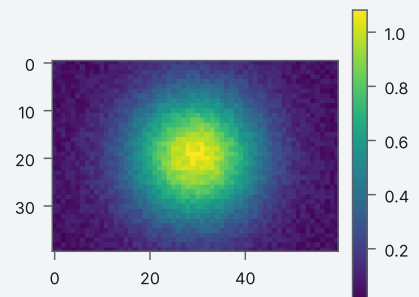


RdBu in light, berlin in dark: a new image

Heatmap, viridis in both

ISO 32000-2 Table 87, T61

stays

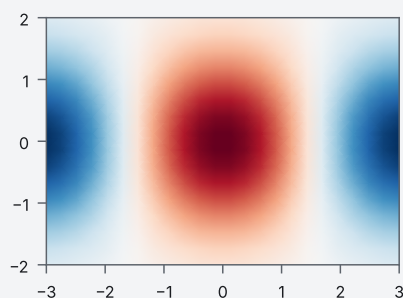


Data colours kept; only the frame swaps

Gouraud mesh (pcolormesh)

ISO 32000-2 Table 81

swaps

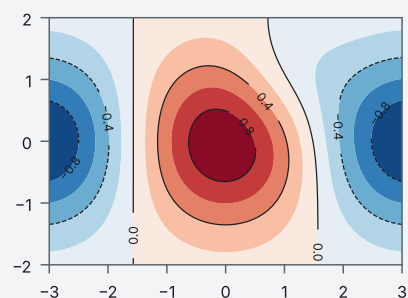


Type 4 shading with a diverging map

Filled and labelled contours

ISO 32000-2 Table 59, T110

swaps

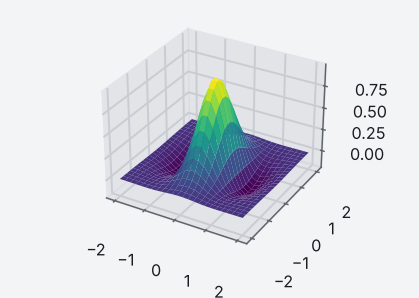


Contour fills swap with the map

3D surface (mplot3d)

ISO 32000-2 Table 59

stays

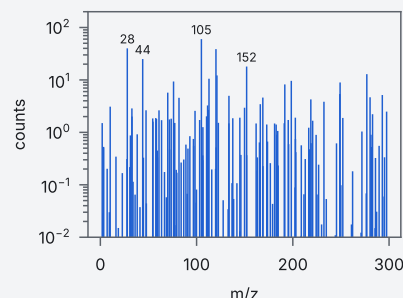


Hundreds of polygons, each its own colour

Mass spectrum: sticks, log scale

ISO 32000-2 Table 59, T110

swaps

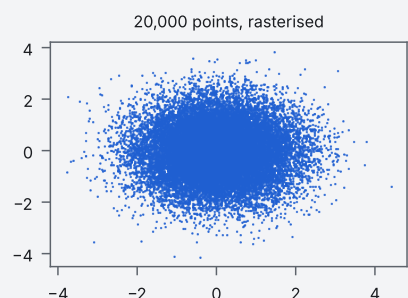


Peak labels and 164 sticks

Rasterised layer in a vector plot

ISO 32000-2 Table 87, T144

swaps

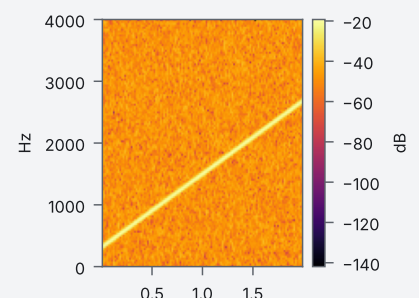


The points are baked into an image

Spectrogram with colour bar

ISO 32000-2 Table 87

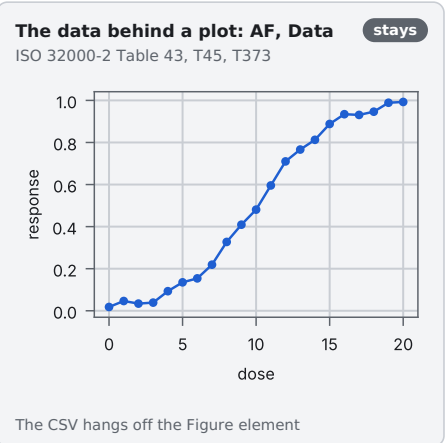
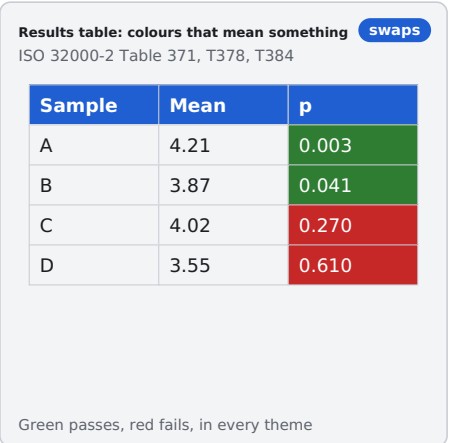
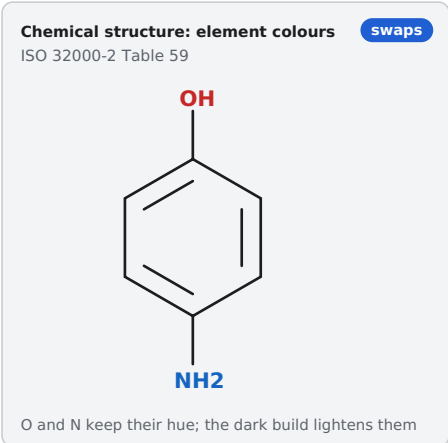
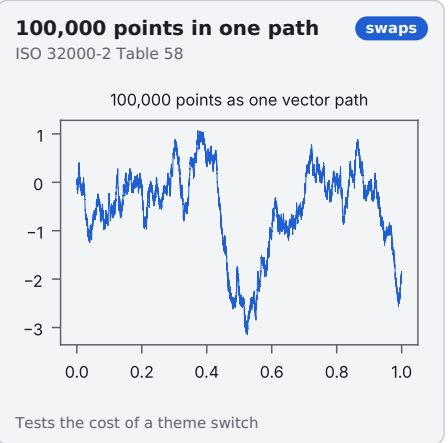
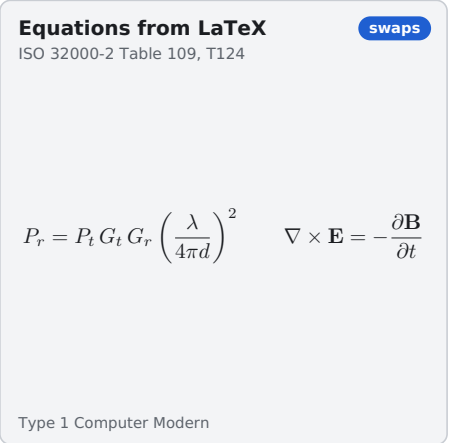
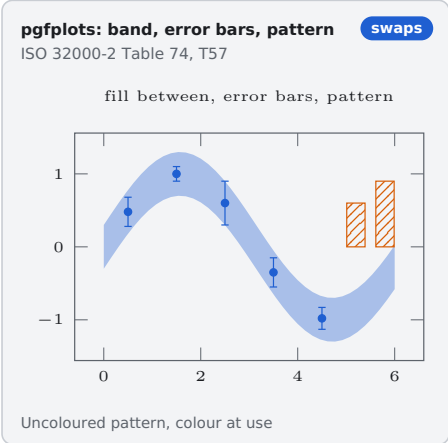
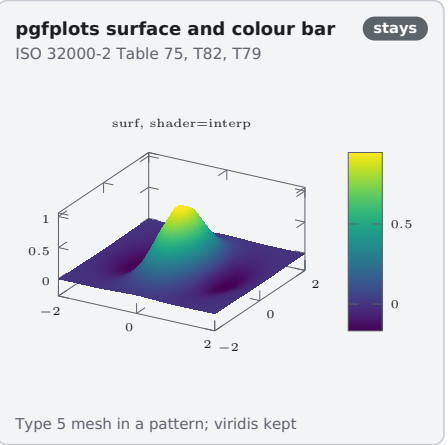
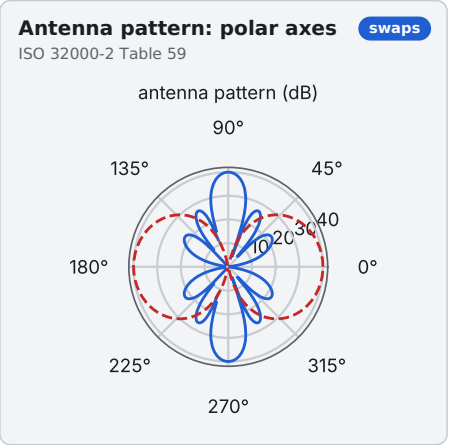
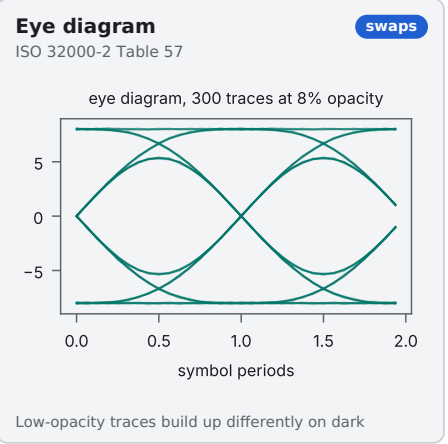
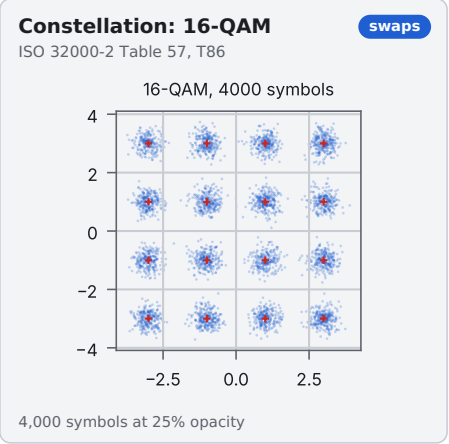
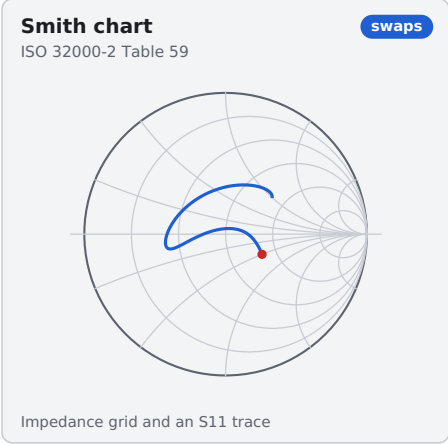
stays



inferno in both builds

26 Scientific figures: RF, imaging, LaTeX and data

RF plots, a false-colour image with a scale bar, pgfplots and LaTeX output, a chemical structure, a results table whose colours mean something, and the data behind a plot attached to the file.



27 Appendix: coverage

Every tile in this file, the ISO 32000-2 tables it exercises, and what a theme should do with it. The table is generated from the same source as the pages.

- swaps** The theme must change it. The dark build shows the intended result.
- stays** Photos, print-production marks and media keep their colours in every theme.
- viewer** Drawn by the viewer from entries in the file, so the viewer has to follow the theme.

Not included, and why

XFA forms	Deprecated in PDF 2.0; most viewers no longer render them.
PostScript XObjects and the PS operator	Removed in PDF 2.0.
Web Capture (SpiderInfo, IDS, URLS)	Records where pages came from; draws nothing.
Alternate presentations (slide shows)	Deprecated; no current viewer plays them.
JBIG2 symbol dictionaries and text regions	Generic regions are tested; symbol coding needs an encoder we don't have.
Portable collections (portfolios)	Change how the whole file opens, so they're a separate test pair.
Encryption and signatures	Change how the file is stored, not what's drawn: separate variant files.
Annotations and fields without appearances	Not allowed in PDF 2.0, but common: they're in the legacy pair.
3D PRC streams	No PRC writer was available; the U3D stream carries the same PDF-side colours.

Page	Object	ISO 32000-2 tables	Theme
1	Fill rules: f and f*	59	swaps
1	Line widths	56, 51	swaps
1	Line caps: J 0, 1, 2	53	swaps
1	Line joins and miter limit	54, 56	swaps
1	Dash patterns	55, 56	swaps
1	Curves: c, v and y	58	swaps
1	Fill and stroke: B, B*, b, b*, s	59	swaps
1	Clipping: W and W*	60	swaps
1	Rectangles, hairlines and dots	58, 51	swaps
1	Graphics state: q, Q, cm and gs	56, 57	swaps
1	Stroke adjustment and flatness	57	viewer
1	Strokes under a scaled CTM	56	swaps
2	DeviceGray: g, G	61, 73	swaps
2	DeviceRGB: rg, RG	61, 73	swaps
2	DeviceCMYK: k, K	61, 73	swaps
2	CalGray	62	swaps
2	CalRGB	63	swaps
2	Lab	64	swaps
2	ICCBased: sRGB	65, 66	swaps
2	ICCBased: Display P3	65, 67	swaps
2	ICCBased: CMYK (FOGRA39)	65, 68	swaps
2	Indexed over RGB, Lab and ICC	61	swaps
2	Separation: spot, All, None	61	swaps
2	DeviceN and NChannel	70, 71	swaps
2	Rendering intents: ri	69	viewer
2	Default colour spaces in a form	34, 61	swaps
2	Overprint: OP, op and OPM	57, 146	viewer
3	Type 1: function-based	78, 42	swaps
3	Type 2: axial	79, 40	swaps
3	Type 3: radial	80, 41	swaps
3	Axial with a sampled function	79, 39	swaps
3	Type 4: free-form triangle mesh	81	swaps
3	Type 5: lattice-form mesh	82	swaps
3	Type 6: Coons patch mesh	83, 84	swaps
3	Type 7: tensor-product patch mesh	83, 85	swaps
3	Mesh with a function	81	swaps
3	Axial shading in Lab	79, 64	swaps
3	Axial shading in DeviceN	79, 70	swaps
3	Background, BBox and AntiAlias	77	swaps
4	Tiling, coloured (PaintType 1)	74	swaps
4	Tiling, uncoloured (PaintType 2)	74	swaps
4	Tiling types 1, 2, 3 and a Matrix	74	swaps
4	Shading pattern: fill	75	swaps
4	Shading pattern: stroke	75	swaps
4	Shading pattern: text	75, 104	swaps
4	Pattern inside a pattern	74	swaps
4	Shading pattern with ExtGState	75, 57	swaps
4	Uncoloured pattern over a spot colour	74, 61	swaps

28 Appendix: coverage, continued

Page	Object	ISO 32000-2 tables	Theme
5	Standard 14 fonts, not embedded	108, 109	swaps
5	Type 1, embedded (FontFile)	109, 124	swaps
5	TrueType, embedded (FontFile2)	109, 124	swaps
5	Bare CFF (FontFile3 /Type1C)	124, 125	swaps
5	OpenType CFF (FontFile3 /OpenType)	124, 125	swaps
5	Type 0 with CIDFontType2, Identity-H	115, 119	swaps
5	Type 0 with CIDFontType0 (CFF CIDs)	115, 119	swaps
5	Vertical writing: Identity-V	115, 116	swaps
5	Text state: Tc, Tw, Tz, Ts, TL, TJ	103, 106, 107	swaps
5	Text matrix: Tm	106	swaps
6	Type 3, uncoloured glyphs (d1)	110, 111	swaps
6	Type 3, coloured glyphs (d0)	110, 111	swaps
6	Type 3 glyph with a bitmap	110	stays
6	Rendering modes 0 to 3: Tr	104	swaps
6	Rendering modes 4 to 7: clip	104	swaps
6	Text knockout: TK	57	swaps
6	Invisible text over a scan	104	stays
6	Fill, stroke and pattern together	104, 75	swaps
6	Tiny, huge and scaled text	103	swaps
7	DeviceGray, 8-bit photo	87, 61	stays
7	DeviceGray at 1, 2 and 4 bits	87, 88	stays
7	DeviceRGB, 8-bit photo	87	stays
7	DeviceRGB, 16-bit samples	87	stays
7	DeviceCMYK photo	87, 61	stays
7	Lab photo	87, 64	stays
7	ICCBased: sRGB and Display P3	87, 65	stays
7	Indexed, 8-bit diagram	87, 61	swaps
7	Indexed at 1, 2 and 4 bits	87, 61	swaps
7	Indexed over Lab and ICC	87, 61	swaps
7	Separation image (spot ink)	87, 61	swaps
7	DeviceN image (two spot inks)	87, 70	swaps
8	FlateDecode	6, 8	stays
8	FlateDecode with PNG predictor	8, 9, 10	stays
8	LZWDecode	6, 7, 8	stays
8	RunLengthDecode	6	swaps
8	ASCIIHexDecode then Flate	6	stays
8	ASCII85Decode	6	stays
8	DCTDecode (JPEG), baseline	6, 13	stays
8	DCTDecode, progressive greyscale	6, 13	stays
8	DCTDecode, CMYK (Adobe)	6, 13, 88	stays
8	JPXDecode (JPEG 2000)	6	stays
8	CCITTFaxDecode, Group 4	6, 11	stays
8	CCITTFaxDecode, Group 3	6, 11	stays
8	JBIG2Decode (generic region, MMR)	6, 12	stays
8	Interpolate: false and true	87	stays
8	Alternates and a Decode array	87, 89, 88	stays
9	Stencil mask (ImageMask)	87, 88	swaps
9	Stencil mask with Decode [1 0]	88	swaps
9	Explicit mask (Mask stream)	87	stays
9	Colour-key mask (Mask array)	87	swaps
9	Soft mask (SMask)	87, 143	stays
9	Soft mask with Matte	144	stays
9	JPEG 2000 with SMaskInData	87	stays
9	Inline images: grey, RGB, Indexed	90, 91, 92	swaps
9	Inline stencil mask	91	swaps
10	Fill opacity: ca	57, 136	swaps
10	Stroke opacity: CA	57	swaps
10	Alpha is shape: AIS	57	swaps
10	All sixteen blend modes: BM	134, 135	swaps
10	Highlighter: Multiply, or Screen in dark	134	swaps
10	Screen and Lighten on paper	134	swaps
10	Difference and Exclusion on text	134	swaps
11	Isolated and knockout groups: I and K	94, 145	swaps
11	Group colour space: CS	145	swaps
11	Luminosity soft mask with BC	142	swaps
11	Alpha soft mask	142	swaps

29 Appendix: coverage, continued

Page	Object	ISO 32000-2 tables	Theme
11	Soft mask with a transfer function	142	swaps
11	SMask None resets the mask	57	swaps
11	Transparency group reused	94	swaps
12	One form, placed four times	93, 86	swaps
12	Nested forms, three deep	93	swaps
12	Form drawn in the caller's colour	93	swaps
12	BBox clips and Matrix maps	93	swaps
12	Form with Measure and PtData (PDF 2.0)	93, 266, 272	swaps
12	Reference XObject (Ref)	93, 95	swaps
12	Form in a tiling pattern cell	74, 93	swaps
12	Form with its own Group and Matrix	93, 94	swaps
12	Image inside a form, reused	93, 87	stays
13	Default on	96, 99	swaps
13	Default off	96, 99	swaps
13	View and print usage	100, 101	swaps
13	Print-only layer	100, 101	swaps
13	Zoom-dependent layer	100	swaps
13	Language layers	100	swaps
13	Radio-button layers: RBGroups	99	swaps
13	Visibility expression: OCMD VE	97	swaps
13	Locked layer, layered annotation and form	99, 166, 93	swaps
14	Text (sticky note) with appearance	175, 186	swaps
14	Review state and a reply: IRT, RT, State	172, 174, 175	swaps
14	FreeText with DA, DS and RC	177	swaps
14	FreeText callout: CL, LE, IT	177, 179	swaps
14	FreeText typewriter	177	swaps
14	Line: all ten line endings	178, 179	swaps
14	Line dimension with Measure	178, 266, 267	swaps
14	Square and Circle with IC	180, 168	swaps
14	Cloudy border: BE	169	swaps
14	Polygon and PolyLine	181	swaps
14	Ink: InkList	185	swaps
14	C and IC in grey, CMYK and none	166, 180	swaps
15	Highlight	182	swaps
15	Underline, Squiggly, StrikeOut	182	swaps
15	Caret: Sy P and None	183	swaps
15	Stamp: drawn and image	184	swaps
15	Stamp names, with appearances	184	swaps
15	Popup, open	186	viewer
15	Redact (marked, not applied)	195	swaps
15	FileAttachment: six icon names	187, 43	swaps
15	Watermark with FixedPrint	193, 194	swaps
15	PrinterMark: colour bar and target	398, 399	stays
15	Annotation BM and ca (PDF 2.0), R and D	166, 170	swaps
15	TrapNet (deprecated in PDF 2.0)	403, 404	stays
16	Link with a border colour: C	176, 210	viewer
16	Highlight modes: H	176	viewer
16	Link over two lines: QuadPoints	176	viewer
16	Link with a dashed border style: BS	168, 176	viewer
16	Destinations: XYZ, Fit, FitH, FitR, FitB	149	swaps
16	Named, remote and embedded go-to	202, 203, 204, 205	swaps
16	Named actions	215, 216	swaps
16	Hide action and the Hidden flag	214, 167	swaps
16	JavaScript action and Next	221, 196	viewer
16	Launch, Thread and Transition actions	207, 209, 219	viewer
16	Flags: NoView, Print off, ToggleNoView	167	viewer
16	Page and document additional actions	197, 198, 200	viewer
17	Text field	226, 228, 232	swaps
17	Multiline text field	231	swaps
17	Password field	231	swaps
17	Comb field: MaxLen 6	231, 232	swaps
17	File-select field	231	swaps
17	Rich text field: RV and DS	228, 231	swaps
17	Alignment: Q 1 and Q 2	228	swaps
17	Border styles: beveled, inset, underline	168, 192	swaps
17	Required and read-only flags	227	swaps

30 Appendix: coverage, continued

Page	Object	ISO 32000-2 tables	Theme
17	Number format script: red negatives	199, 221	viewer
17	Calculated field: CO	199, 224	viewer
17	List box	233, 234	swaps
17	Multi-select list box	233, 234	swaps
17	Combo box	233	swaps
17	Editable combo box	233	swaps
18	Check boxes: six styles	229, 230, 192	swaps
18	Radio group	229, 230	swaps
18	Push button: N, R and D appearances	229, 170, 192, 241	swaps
18	Icon buttons: MK I, RI, IX, IF and TP	192, 250	swaps
18	Submit button	239, 240	viewer
18	Signature field, unsigned	235, 236, 237	swaps
18	Focus and blur scripts recolour a field	197, 221	viewer
18	Hidden field shown by a button	167, 214	swaps
18	Field hierarchy: a.b names	226	swaps
18	Read-only check box and NoExport	227	swaps
18	Import data action	243	viewer
18	MK colours in grey and CMYK	192	swaps
19	Screen with a Rendition action	190, 218, 282, 293	swaps
19	Rendition actions: play, pause, stop	218	viewer
19	RichMedia (video)	333, 334, 335, 338, 339, 340, 341, 342, 222	swaps
19	Movie annotation (deprecated)	189, 306, 307, 213	swaps
19	Sound annotation and action (deprecated)	188, 305, 212	swaps
19	3D annotation (U3D)	309, 310, 311, 315, 317, 318, 320, 322, 323, 326-331	swaps
19	GoTo3DView actions, 3D markup	220, 173, 324	viewer
19	Projection annotation (PDF 2.0)	172, 332, 327	swaps
19	Selector rendition with MH and BE	277, 278, 279, 283	viewer
19	Floating window colour	293, 294, 295	swaps
19	Embedded media files	43, 44, 45	stays
20	Page boxes and BoxColorInfo	31, 396, 397	viewer
20	Thumbnail: Thumb	31	swaps
20	Transition and display time	31, 164	viewer
20	Article thread: beads	162, 163	viewer
20	Viewport with a scale: Measure RL	265, 266, 267, 268	swaps
20	Geospatial viewport: GEO, PROJCS, PtData	265, 269, 270, 271, 272	swaps
20	Navigation steps: PresSteps	165	viewer
20	Document parts and GoToDp	206, 408, 409	swaps
20	Associated file on a page: AF	31, 43, 45	stays
20	Outlines with colour and style	151, 152	swaps
20	Page labels	161	viewer
20	Page output intent, by reference	31, 401, 402	stays
21	Content on a rotated page	31	swaps
21	NoRotate and NoZoom flags	167	viewer
21	Article thread continues	163	viewer
21	Last bead	163	viewer
22	UserUnit page	31	swaps
23	Table with Layout colours	371, 377, 378, 384	swaps
23	List with numbering	370, 382	swaps
23	Underline with TextDecorationColor	380	swaps
23	Attribute class: ClassMap and C	354, 355, 360	swaps
23	FENote, E and Lang (PDF 2.0 namespace)	355, 356, 366, 368	swaps
23	Formula with a MathML file	374, 43	swaps
23	Ruby and Warichu	369	swaps
23	Figure with BBox and Caption	373, 372, 379	swaps
24	Halftone type 1: spot functions	126, 127, 128	stays
24	Halftone type 5 per colorant	132	stays
24	Threshold halftones: types 6, 10, 16	129, 130, 131	stays
24	Transfer functions: TR and TR2	57	viewer
24	Black generation and undercolour removal	57	stays
24	SM, HTO and UseBlackPtComp	57	stays
24	SeparationInfo	400	stays
24	Document output intent	29, 401	stays
24	OPI 1.3 and 2.0 on an image (deprecated)	405, 406, 407	stays
24	MMType1 font (deprecated)	108, 109	swaps
24	BX/EX and marked points: MP, DP	33, 352	swaps
25	Line plot: series, legend, grid	59, 110	swaps

31 Appendix: coverage, continued

Page	Object	ISO 32000-2 tables	Theme
25	Scatter coloured by value	86, 93, 87	swaps
25	Error bars and a confidence band	57, 86	swaps
25	Histogram with hatching	74, 57	swaps
25	Heatmap, diverging map	87, 61	swaps
25	Heatmap, viridis in both	87, 61	stays
25	Gouraud mesh (pcolormesh)	81	swaps
25	Filled and labelled contours	59, 110	swaps
25	3D surface (mplot3d)	59	stays
25	Mass spectrum: sticks, log scale	59, 110	swaps
25	Rasterised layer in a vector plot	87, 144	swaps
25	Spectrogram with colour bar	87	stays
26	Smith chart	59	swaps
26	Constellation: 16-QAM	57, 86	swaps
26	Eye diagram	57	swaps
26	Ion image with a scale bar	87	stays
26	Antenna pattern: polar axes	59	swaps
26	pgfplots surface and colour bar	75, 82, 79	stays
26	pgfplots: band, error bars, pattern	74, 57	swaps
26	Equations from LaTeX	109, 124	swaps
26	100,000 points in one path	58	swaps
26	Chemical structure: element colours	59	swaps
26	Results table: colours that mean something	371, 378, 384	swaps
26	The data behind a plot: AF, Data	43, 45, 373	stays